



Sharan (L) and Shashwath (R), Co-founders, Mindgrove Technologies

signing chips for machine learning and characterisation-focused signal processing and vision, to deliver the required performance without resorting to excessive complexity or gate counts, resulting in power efficiency. “Our primary emphasis is on optimising performance, and then we progressively eliminate components and features until we achieve the desired power consumption. We aim to create a chip perfectly suited for its intended purpose, effectively a ‘right-sized’ chip for the task at hand,” Sharan explains.

Operating at 700MHz, the SoC has 128kB of on-chip SRAM, with two quad SPI ports that can operate at 200Mbps each. These ports can be connected to various memory types, including NOR flash, PS RAM

(pseudo static RAM), and M RAM. The chip can accommodate setups like a 64MB NOR flash with one PS RAM, two NOR flashes, or one 64MB NOR flash alongside a secondary NAND flash on a quad SPI interface. Chief Technology Officer (CTO) Sharan says that a versatile memory configuration approach allows the chip to cater to various application needs, while optimising performance and memory usage.

“Our quad SPI ports provide the flexibility to connect various external modules, such as Wi-Fi, Bluetooth, etc, as peripherals to cater to specific customer needs instead of incorporating all these features within the SoC. By not embedding a fixed amount of memory directly into the package, we avoid the

scenario where customers pay for more memory than their use case requires, eliminating unnecessary costs and allowing us to serve a diverse and fragmented market,” he elaborates.

The startup employs BlueSpec SystemVerilog for RTL design and uses a combination of tools for verification followed by a continuous integration approach for testing. It also develops firmware, drivers, and software parallel to hardware development to ensure smooth integration and functionality. “Ventilator is the fastest and open source simulator, making it cost-effective. We use a Python-based ‘Cocotb’ framework to control the data flow. The BlueSpec code gets compiled, the compiled code goes into Ventilator, and Cocotb controls it. We write all our test benches in Cocotb, which is in Python,” explains Sharan.

Shashwath adds that around 85% of the chip was either developed in-house or at IIT Madras, along with IP support from Synopsys. Currently, the startup expects production to start in November 2023. “We anticipate the prototype to be ready by February, after which we will soon bring the product to market. We’re already working on our second project, focusing on image processing and computer vision for integration into cameras and similar devices for on-edge applications,” Shashwath concluded.

—Yashasvini Razdan

3 MOST ENERGY-DENSE ELECTRIC VEHICLE BATTERIES

Apart from cells and chips, the startup has absolute control over the components and raw materials it sources for manufacturing batteries and battery swap stations. Reddy says his startup is focusing on the B2B space as most of the B2C electric vehicle users commute for a limited number of kilometres every day

“Our electric vehicle batteries come packed with almost 170Wh per kg of energy density,” Arun Sreyas Reddy, Founder of RACEnergy, shared in an exclusive conversation with